

Computer Science & IT

Discrete Mathematics



Set Theory & Algebra

Lecture No. 08



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Recap of Previous Lecture



Topic

GATE PYQ



Topic

Partial order relation



Topic

Partially ordered set (POSET)
(A, R)



Topic

Comparability



Topic

Totally ordered set (TOSET)



Topics to be Covered



Topic

Least upper bound (LUB)



Topic

Greatest lower bound (GLB)



Topic

Minimal and maximal elements in a POSET



Topic

Minimum (Least) element of POSET



Topic

Maximum (Greatest) element of POSET



Note:

Let (A, R) be a POSET

And element $a, b \in A$

and $a^R b$ i.e. $(a, b) \in R$ { i.e. elements a & b are Comparable }

'a' is the
element at
the lower side

'b' is the
element at the
upper side



Topic : Least Upper Bound

lub

Join

Supremum



Let $(A, \leq)$ be a POSET, for any two elements $a, b \in A$ if there exists an element $c \in A$ such that,

$a \leq c$ and $b \leq c$,

then c is called upper bound of a and b ,

$(a, c) \in \leq$ & $(b, c) \in \leq$

'c' is at upper side of both a & b

And if there exists no element $d \in A$ such that
($d \neq c$)

$a \leq d$ and $b \leq d$ and $d \leq c$,

$(a, d) \in \leq$ & $(b, d) \in \leq$ & $(d, c) \in \leq$

then c is called least upper bound of a and b

'd' is upper bound of a & b

'd' is at lower side of c



Topic : Least Upper Bound

lub / Join / Supremum



any partial order $\leq$ on set A

Let $(A, \leq)$ be a POSET, for any two elements $a, b \in A$ if there exists an element $c \in A$ such that,

$a \leq c$ and $b \leq c$,

then c is called upper bound of a and b ,

$(a, c) \in \leq$ & $(b, c) \in \leq$

' c ' is at upper side of both a & b

And if there exists no element $d \in A$ such that
($d \neq c$)

$a \leq d$ and $b \leq d$ and $d \leq c$,

then c is called least upper bound of a and b

$(a, d) \in \leq$ & $(b, d) \in \leq$ & $(d, c) \in \leq$

' d ' is upper bound of a & b

' d ' is at lower side of c

* Least upper bounds of two elements a & b
Can be denoted by $\text{lub}(a,b)$ or $a \vee b$

$$\text{lub}(a,b) = \text{lub}(b,a)$$

i.e. $a \vee b = b \vee a$

Notation for
least upper bound



Topic : Greatest Lower Bound

/ glb / Meet / Infimum 

Let $(A, \leq)$ be a POSET, for any two elements $a, b \in A$ if there exists an element $c \in A$ such that,

$c \leq a$ and $c \leq b$,

then c is called lower bound of a and b ,

$(c, a) \in \leq$ & $(c, b) \in \leq$

' c ' is at lower side of both a & b

And if there exists no element $d \in A$ such that

$(d \neq c)$

$d \leq a$ and $d \leq b$ and $c \leq d$,

$(d, a) \in \leq$ & $(d, b) \in \leq$ & $(c, d) \in \leq$

' d ' is lower bound of a & b

' d ' is at upper side of c

then c is called greatest lower bound of a and b



Topic : Greatest Lower Bound

/ glb / Meet / Infimum 

Partial order R

Let (A, R) be a POSET, for any two elements $a, b \in A$ if there exists an element $c \in A$ such that,

$c R a$ and $c R b$,

then c is called lower bound of a and b ,

$(c, a) \in R$ & $(c, b) \in R$

'c' is at lower side of both a & b

And if there exists no element $d \in A$ such that $(d \neq c)$

$d R a$ and $d R b$ and $c R d$,

$(d, a) \in R$ & $(d, b) \in R$ & $(c, d) \in R$

then c is called greatest lower bound of a and b

'd' is lower bound of a & b

'd' is at upper side of c

Note:- Greatest lower bound of two elements a & b
Can be denoted by $\text{glb}(a, b)$ or $a \wedge b$

↖ Notation
for glb



$$\text{glb}(a, b) = \text{glb}(b, a)$$

i.e. $a \wedge b = b \wedge a$

eg: Let $A = \{1, 2, 3, 4\}$, and $(A, \leq)$ is a POSET

then: $\text{lub}(2, 3) = ?$ & $\text{glb}(2, 3) = ?$

$2 \not\leq 1$	$2 \leq 2$	$2 \leq 3$	$2 \leq 4$
$3 \not\leq 1$	$3 \not\leq 2$	$3 \leq 3$	$3 \leq 4$

Elements 3 & 4 are
at upper side of
both '2' & '3'
 $\therefore$ '3' & '4' both are
upper bounds of '2' & '3'

$\text{lub}(2, 3) = 3$

But $3 \leq 4$, i.e. '3' is at
lower side of '4' Hence '4' cannot
be the lub of 2 & 3

$1 \leq 2$	$2 \leq 2$
$1 \leq 3$	$2 \leq 3$

Elements 1 & 2 are
at lower side of
both '2' & '3'
 $\therefore$ '1' & '2' both are
lower bounds of 2 & 3

But $1 \leq 2$, i.e. '2' is at
upper side of '1' Hence '1'
cannot be the glb of 2 & 3

$3 \not\leq 2$	$4 \not\leq 2$
$3 \leq 3$	$4 \not\leq 3$

$\therefore \text{glb}(2, 3) = 2$

eg. Let $A = \{1, 2, 3, 4\}$
and $(A, \geq)$ is a POSET,

$$\text{lub}(2, 3) = ?$$

$$\text{glb}(2, 3) = ?$$

Similarly $\text{glb}(2, 3) = 3$

$2 \geq 1$	$2 \geq 2$	$2 \not\geq 3$	$2 \not\geq 4$
$(2, 1) \in \geq$	$(2, 2) \in \geq$		
$3 \geq 1$	$3 \geq 2$	$3 \geq 3$	$3 \not\geq 4$
$(3, 1) \in \geq$	$(3, 2) \in \geq$	$(3, 3) \in \geq$	

1 & 2 are at upper side
of both 2 & 3 $\therefore$ 1 & 2 are upper bounds of 2 & 3

But $(2, 1) \in \geq$ i.e., '2' is at lower side of '1', Hence '1' cannot be the lub of 2 & 3
 $\text{lub}(2, 3) = 2$

Note:- If A is any set of real numbers and $(A, \leq)$ is a POSET, then for two elements $a, b \in A$

$$\text{lub}(a, b) = \text{Max}(a, b)$$

$$\text{glb}(a, b) = \text{Min}(a, b)$$

Note : $\rightarrow$

$(a, b) \in \div$
iff 'a' divides 'b'
ie. if 'a' divides 'b'
then 'a' is at
lower side
&
'b' is at
upper side

Let A is a set of all positive integers
and $(A, \div)$ is a POSET.

for two elements $a, b \in A$

$$\text{lub}(a, b) = \text{LCM}(a, b)$$

$$\text{glb}(a, b) = \text{GCD}(a, b)$$

Note: →

if $A \subseteq B$.
then $(A, B) \in \subseteq$

i.e. if $A \subseteq B$, then
A at lower side
&
B at upper side

Let A is any finite set and $P(A)$ is the power set of set A , then $(P(A), \subseteq)$ is a POSET.

For two elements (sets) $X, Y \in P(A)$

$\text{lub}(X, Y) = X \text{ Union } Y$

$\text{glb}(X, Y) = X \text{ intersection } Y$

Note:- If A is any set of Positive integers
and $(A, \div)$ is a POSET.

→ then for two elements $a, b \in A$

$\text{lub}(a, b) =$ Nothing can be said
Because we don't know the elements of the set

$\text{glb}(a, b) =$ Nothing can be said
Because we don't know the elements of the set

Note:- Let C is any Collection of sets, then
 $(C, \subseteq)$ is a POSET

for two sets $X, Y \in C$

$\text{lub}(X, Y) =$ Nothing Can be said
We don't know the elements of set C

$\text{glb}(X, Y) =$ Nothing Can be said
We don't know the elements of set C

eg. $C = \{ \{1\}, \{2\}, \{1, 2, 3, 4, 5\} \}$

$(C, \subseteq)$ is a POSET

$$\text{lub}(\{1\}, \{2\}) = \{1, 2, 3, 4, 5\}$$

$$\text{glb}(\{1\}, \{2\}) = \text{Does not exist}$$

eg. $C = \{ \{1\}, \{2\}, \{1, 2, 3\}, \{1, 2, 4, 5\} \}$

$(C, \subseteq)$ is a POSET

Both $\{1, 2, 3\}$ & $\{1, 2, 4, 5\}$ are upper bounds of $\{1\}$ & $\{2\}$
 $\{1, 2, 4, 5\} \not\subseteq \{1, 2, 3\}$ & $\{1, 2, 3\} \not\subseteq \{1, 2, 4, 5\}$

$\text{lub}(\{1\}, \{2\}) =$

W.r.t. some authors,
both $\{1, 2, 3\}$ & $\{1, 2, 4, 5\}$ are lub of $\{1\}$ & $\{2\}$

W.r.t. some authors,
lub of $\{1\}$ & $\{2\}$ does not exist

eg. $A = \{ 2, 3, 12, 24 \}$

and $(A, \div)$ is a POSET

then, $\text{lub}(2, 3) = 12$

$$\text{glb}(2, 3) = \top \text{ N E}$$

Note :- →

In a POSET lub and/or glb
need not exist for Every pair
of elements



Topic : Minimal elements in a POSET

R ← it can be any Partial order R

More than one are also possible

Let $(A, \leq)$ be a POSET

An element $x \in A$ is called a minimal element of $\text{POSET}(A, \leq)$ if there exists no element $y \in A$ other than x itself, such that $y \leq x$. i.e. $(y, x) \in \leq^{(A, R)}$

i.e. $(y, x) \in R$

at lower side of x



Topic : Minimal elements in a POSET

Let $(A, \leq)$ be a POSET

An element $x \in A$ is called a minimal element of $\text{POSET}(A, \leq)$ if there exists no element $y \in A$ other than x itself, such that $y \leq x$.

ie., if there is no element 'y' other than 'x' itself
which is at the lower side of x, then
'x' is a minimal element of the POSET.

eg:- $A = \{2, 3, 4, 6, 12\}$
and $(A, \div)$ is a POSET.

find all minimal elements in the POSET.

2 & 3 are minimal elements.

$$\begin{aligned} (2, 4) &\in \div \\ (2, 6) &\in \div \\ (4, 12) &\in \div \end{aligned}$$

$\therefore 4, 6 \& 12$ are
not minimal



Topic : Maximal elements in a POSET

Any Partial order
 R

Let $(A, \leq)$ be a POSET

An element $x \in A$ is called a maximal element of $\text{POSET}(A, \leq)$ if there exists
no element $y \in A$ other than x itself, such that $x \leq y$.

$(x, y) \in R$

upper side of x



Topic : Maximal elements in a POSET

Let $(A, \leq)$ be a POSET

An element $x \in A$ is called a maximal element of $\text{POSET}(A, \leq)$ if there exists no element $y \in A$ other than x itself, such that $x \leq y$.

If there is no element 'y', other than 'x' itself
Which is at the upper side of x, then
x is the maximal element of POSET.

eg:- $A = \{2, 3, 4, 6, 15\}$

and $(A, \div)$ is a POSET.

find all maximal elements in the POSET.

$(2, 4) \in \div$

$(3, 15) \in \div$

Not Maximal

4, 6 & 15 are maximal elements of POSET

eg:- $A = \{2, 3, 4, 6, 12\}$

and $(A, \div)$ is a POSET.

find all maximal and minimal elements of the POSET

Maximal element = 12 only

Minimal element = 2 & 3

eg:- $A = \{1, 2, 3, 4, 6, 12\}$
and $(A, \div)$ is a POSET.

Find all minimal and maximal elements of POSET.

Maximal element = 12 only

Minimal element = 1 only



Topic : Minimum element of a POSET

Let (A, R) be a POSET

An element which is at the lower side of all the elements of the set is called minimum element

An element $m \in A$ is called a minimum element of $\text{POSET}(A, R)$

if and only if

$$m R a, \forall a \in A$$

ie $(m, a) \in R, \forall a \in A$

'm' is at lower side of

all elements of set A

eg:- $A = \{2, 3, 4, 6, 15\}$

and $(A, \div)$ is a POSET.

Find minimum element of POSET.

No element of the set divides all elements of the set.
∴ Minimum element does not exist in above POSET.

In above POSET
2 & 3 both are minimal elements.

Note:-

If there exists two or more minimal elements in a POSET, then minimum element can never exist in that POSET.

eg:- $A = \{1, 2, 3, 4, 6, 12\}$
and $(A, \div)$ is a POSET.

find minimum element of POSET.

'1' divides all $\Rightarrow \therefore$ '1' is minimum element

'1' is also the Unique minimal element

Note:-

If there exists a unique minimal element in the POSET, then that element will also be the minimum element of POSET.

Note:

Minimum element if exists, then it is always unique.



Topic : Maximum element of a POSET

↳ An element which is at the upper side of all elements of POSET, is called maximum element.

Let (A, R) be a POSET

An element $n \in A$ is called a maximum element of $\text{POSET}(A, R)$

if and only if

$a R n, \forall a \in A$

i.e. $(a, n) \in R, \forall a \in A$

'n' is at
upper side
of

all elements
'a' of set A

eg:- $A = \{2, 3, 4, 6, 15\}$

and $(A, \div)$ is a POSET.

Find maximum element of POSET.

Maximum element does not exist

But 4, 6 & 15 are maximal elements of POSET.

Note:-

If there exists two or more maximal elements in a POSET, then maximum element can never exist in that POSET.

eg:- $A = \{1, 2, 3, 4, 6, 12\}$

and $(A, \div)$ is a POSET.

Find maximum element of POSET.

'12' is divided by all

or Every element of set A divides '12' $\Rightarrow \therefore$ '12' is maximum element.

'12' is also the Unique maximal element

Note:- If there exists a unique maximal element in the POSET, then that element will also be the maximum element of POSET.

Note: Maximum element if exists, then it is always unique.

Note:-

In a POSET, minimum and/or maximum element need not exist, but if exists then they will be unique.



Topic : NOTE



Let Z is the set of all integers, then
 $(Z, \leq)$ is a POSET.

$(-\infty, \infty)$

No minimal element exist in the POSET.

Hence No minimum as well

No maximal element exist in the POSET.

Hence No Maximum as well.



Topic : NOTE



- Let N is the set of all natural numbers, then $(N, \leq)$ is a POSET.

let $[1, \infty)$ ←

- '1' is the only minimal element of POSET
and hence '1' is also the minimum element

- No maximal element exist in the POSET.
Hence No maximum as well.



Topic : NOTE



eg. Let $A = \{x \mid x \in \mathbb{R} \text{ and } 0 \leq x \leq 1\}$

infinite set.

and $(A, \leq)$ is a POSET.

↳ When, '0' is the minimal as well as minimum element

'1' is the maximal as well as maximum element

→ If there is no minimal element in the POSET and/or no maximal element in the POSET, then Set is guaranteed to be an infinite set.

→ But, if Set is an infinite set then minimal and/or maximal elements may or may not exist.

Note :- ① If set is a finite set, then in
a POSET w.r.t. that set.

there will be at least one minimal element
and there will be at least one maximal element

→ ② In a POSET { may or may not be finite }
there can be at most one minimum element
and there can be at most one maximum element



2 mins Summary



✓ **Topic**

Least upper bound (LUB)

✓ **Topic**

Greatest lower bound (GLB)

✓ **Topic**

Minimal and maximal elements in a POSET

✓ **Topic**

Minimum (Least) element of POSET

✓ **Topic**

Maximum (Greatest) element of POSET

THANK - YOU